

The Utility Problem

Outages and reliability are one of the most important metrics for utility companies. Electricity outages have three significant negative impacts on electricity distribution utilities: customers become dissatisfied with their service, regulators impose restrictions and fines for frequent and long-duration outages, and utilities lose revenue due to unused energy during the outage. Although there are different stakeholders for each of the three ramifications, these are all significant concerns for the utility as a whole.

For these reasons, it is necessary to eliminate the outage as soon as possible. For fast recovery, among other things, a comprehensive visualization is necessary. The outage must be shown directly on the map so that the dispatcher and the field worker know precisely where the outage has occurred. The map must also show the affected area with the number of customers affected. If multiple outages occur, the repair of the outages must be prioritized, which is only possible with this information. It is also essential to know the classifications of customers affected to prioritize repairs for important customers (e.g. hospitals, furnaces, etc.).

Detailed information about the location of the outage can also help repair the outage quickly. Therefore, it is necessary to have information not only from the Outage Management System but also from other systems like the Client Information System (CIS) and/or Geographical Information System (GIS), which will pinpoint the location and component damaged for the field crew, removing the need for manual inspection.

Proper visualization and evaluation of the outage can help to reduce the reliability indicators (i.e. SAIDI and SAIFI), which will lead to bonuses for the utility and therefore increase the utility's profits.

The Data Solutions and Methods

The Awesense Platform is capable of this analysis. To perform such visualization, two types of data are required:

1. Outage Management System (OMS) data.
 - a. Enables the utility to identify the location of the outage and/or the set of affected consumers.
 - b. Further analysis using this data can allow for predictions of future outages.
2. Geospatial Information System (GIS) and Connectivity data.
 - a. Along with Awesense's Energy Transition Platform, utilities can, for example, trace down the feeder to identify which meters are experiencing outages.

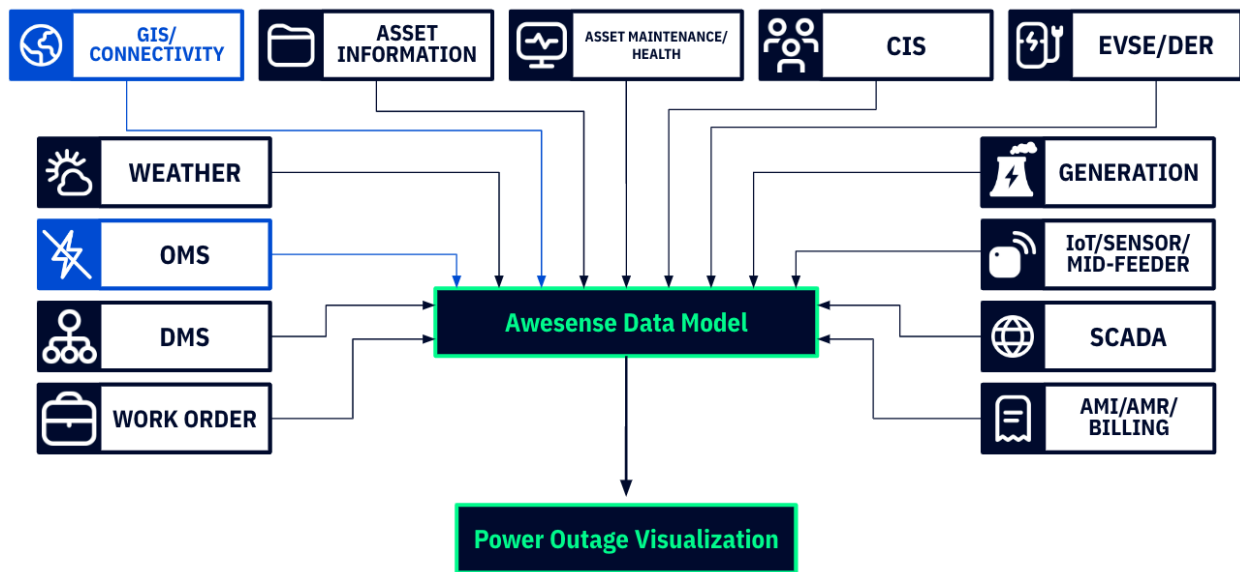
Awesense True Grid Intelligence (TGI) frontend visualization application has the Outages page. There you can view all outages - not only current but also historical (together with a timestamp, when the outage occurred, how many customers were affected, on which element the failure occurred, etc.). In TGI, the data is displayed in a table and on a map. The map base, together

with all the elements of the distribution grid, enables users to immediately see which part of the grid and which location is involved and therefore, all the necessary visualizations can be produced.

After the outages are generated and stored within the Energy Data Model, the Awesense Platform offers a very easy way to perform any additional geospatial analysis for outage areas. With the accurate topological data, it is possible to create analyses for any large part of the grid and drill into the details needed for such analyses.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in royal blue are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. This use case is implemented in TGI as part of the Power Outage Visualization functionality. Consult [instructions](#) about how to access this use case in TGI. If you do not have access to a server where the use case functionality is available, contact [Awesense](#) to see a demo of it.